

# OCD Patient Analytics using Python, SQL Server, and Power BI

## 1. Project Overview

This project analyses healthcare data of obsessive-compulsive disorder (OCD) patients using clinical and behavioural records. The goal is to uncover insights into patient demographics, OCD diagnosis trends, obsession and compulsion patterns, and symptom severity to support better healthcare analysis and decision-making.

## 2. Dataset Summary

- Rows: 1500
- Columns: 17
- Key features:
  - Patient demographics (Age, Gender, Ethnicity, Marital Status, Education Level)
  - Clinical details (OCD Diagnosis Date, Obsession Type, Compulsion Type, Family History of OCD)
  - Behavioural and treatment information (Y-BOCS Obsession Score, Y-BOCS Compulsion Score, Depression Diagnosis, Anxiety Diagnosis, Medication Taken)

## 3. Exploratory Data Analysis Using Python

We began with data preparation and cleaning in python:

- **Data Loading:** Imported the dataset using `pandas`
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

|       | Patient ID  | Age         | Duration of Symptoms (months) | Y-BOCS Score (Obsessions) | Y-BOCS Score (Compulsions) |
|-------|-------------|-------------|-------------------------------|---------------------------|----------------------------|
| count | 1500.000000 | 1500.000000 | 1500.000000                   | 1500.000000               | 1500.000000                |
| mean  | 5541.254000 | 46.781333   | 121.745333                    | 20.048000                 | 19.62600                   |
| std   | 2562.389469 | 16.830321   | 67.404610                     | 11.823884                 | 11.78287                   |
| min   | 1017.000000 | 18.000000   | 6.000000                      | 0.000000                  | 0.00000                    |
| 25%   | 3338.000000 | 32.000000   | 64.000000                     | 10.000000                 | 9.00000                    |
| 50%   | 5539.500000 | 47.000000   | 121.000000                    | 20.000000                 | 20.00000                   |
| 75%   | 7745.500000 | 61.000000   | 178.000000                    | 31.000000                 | 29.00000                   |
| max   | 9995.000000 | 75.000000   | 240.000000                    | 40.000000                 | 40.00000                   |

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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 17 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Patient ID                               1500 non-null   int64
1   Age                                       1500 non-null   int64
2   Gender                                   1500 non-null   object
3   Ethnicity                               1500 non-null   object
4   Marital Status                           1500 non-null   object
5   Education Level                           1500 non-null   object
6   OCD Diagnosis Date                       1500 non-null   object
7   Duration of Symptoms (months)           1500 non-null   int64
8   Previous Diagnoses                       1500 non-null   object
9   Family History of OCD                   1500 non-null   object
10  Obsession Type                           1500 non-null   object
11  Compulsion Type                           1500 non-null   object
12  Y-BOCS Score (Obsessions)                1500 non-null   int64
13  Y-BOCS Score (Compulsions)              1500 non-null   int64
14  Depression Diagnosis                     1500 non-null   object
15  Anxiety Diagnosis                       1500 non-null   object
16  Medications                             1500 non-null   object
dtypes: int64(5), object(12)
memory usage: 199.3+ KB

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- **Data Quality Check:** Verified the dataset for missing values and inconsistencies, no null values were found.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Database Integration:** Connected Python script to Microsoft SQL Server and loaded the cleaned Data frame into the database for SQL analysis.

## 4. Data Analysis Using SQL

We performed structured analysis in Microsoft SQL Server to answer key healthcare and patient-related questions:

- 1) **Gender Wise Distribution of OCD Patients:** Compare the count and percentage of male and female patients along with their average obsession score.

|   | male_count | female_count | pct_male | pct_female | male_avg_obsession_score | female_avg_obsession_score |
|---|------------|--------------|----------|------------|--------------------------|----------------------------|
| 1 | 753        | 747          | 50.20    | 49.80      | 19.90                    | 20.20                      |

- 2) **Ethnicity Based Analysis:** Analyse patient distribution across different ethnicities and compare their average obsession score.

|   | ethnicity | patient_count | avg_obsession_score |
|---|-----------|---------------|---------------------|
| 1 | African   | 324           | 19.76               |
| 2 | Asian     | 386           | 20.32               |
| 3 | Hispanic  | 392           | 20.28               |
| 4 | Caucasian | 398           | 19.79               |

- 3) **Monthly OCD Diagnosis Trend:** Track number of OCD diagnosis month over month to identify diagnosis patterns over time.

|    | month_start | patient_count |
|----|-------------|---------------|
| 1  | 2013-11-01  | 9             |
| 2  | 2013-12-01  | 9             |
| 3  | 2014-01-01  | 13            |
| 4  | 2014-02-01  | 13            |
| 5  | 2014-03-01  | 15            |
| 6  | 2014-04-01  | 12            |
| 7  | 2014-05-01  | 11            |
| 8  | 2014-06-01  | 9             |
| 9  | 2014-07-01  | 11            |
| 10 | 2014-08-01  | 10            |
| 11 | 2014-09-01  | 12            |
| 12 | 2014-10-01  | 19            |
| 13 | 2014-11-01  | 18            |
| 14 | 2014-12-01  | 14            |
| 15 | 2015-01-01  | 16            |
| 16 | 2015-02-01  | 10            |
| 17 | 2015-03-01  | 12            |

- 4) **Most Common Obsession Types:** Identify the most common obsession categories and respective average obsession scores.

|   | obsession_type | patient_count | avg_obsession_score |
|---|----------------|---------------|---------------------|
| 1 | Harm-related   | 333           | 20.65               |
| 2 | Contamination  | 306           | 19.67               |
| 3 | Religious      | 303           | 19.23               |
| 4 | Symmetry       | 280           | 19.67               |
| 5 | Hoarding       | 278           | 21.01               |

- 5) Most Common Compulsion Types: Identify the most common compulsion categories and their respective average obsession scores.

|   | compulsion_type | patient_count | avg_obsession_score |
|---|-----------------|---------------|---------------------|
| 1 | Washing         | 321           | 19.40               |
| 2 | Counting        | 316           | 20.41               |
| 3 | Checking        | 292           | 19.99               |
| 4 | Praying         | 286           | 20.08               |
| 5 | Ordering        | 285           | 20.40               |

## 5. Dashboard in Power BI

Finally, we build a dashboard in **Power BI** to present the insights visually.

